

The Temporal Support Behind One Benchmark Timestamp: A Released Event Detector and DSEC’s Ceiling Exposures

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Abstract

A detection benchmark compares a prediction with a label at one query time and scores their difference as displacement. That timestamp does not uniquely specify the temporal support of either side of the comparison. Event-detection benchmarks collapse three quantities into it: a predictor’s evidence support, its effective output time, and the exposure-conditioned support of the labels. This paper separates the three and measures what released artifacts identify of each. The released configuration of the detector examined here ends its input window at the label instant, and occluding each bin of the released `rvt-t` checkpoint’s input puts that window’s influence centroid at -23.81 ms. The detector is recurrent, so that window is not all of its evidence: occluding whole past windows leaves the newest carrying 22.7 % of the influence, with a tail the measured horizon does not resolve. Its detections, regressed on the label velocity as a vector alongside that velocity rotated a quarter turn as a placebo no temporal offset can reach, put the output time at -2.40 ± 7.18 ms: consistent with the label instant, which its training objective encourages, and 3.65 standard errors from the window’s own centroid. This establishes the separation, not an equality with the label timestamp. That offset spans -20.2 to $+11.1$ ms across specifications with geometric controls, never reaching the $+23.81$ ms the centroid would require. Displacing the ground truth across that interval moves mAP by 0.04 points, so a score that insensitive cannot be used to ask which instant a prediction represents. On the label side no per-object label time is inferred: the six DSEC sequences pinned at the auto-exposure ceiling carry the harmonic series of a 100.00 Hz mains flicker, undocumented in the release, and what follows is an attribution limit. A per-object temporal statistic measured inside one of those exposures cannot be attributed to object timing without separating the illumination-driven structure first.

1 Introduction

An object detection benchmark evaluates a prediction $\hat{s}$ against a ground truth s^* through a function $\rho(\hat{s}, s^*)$ of position, shape and class, indexed by a query time t . The query time selects which label the prediction is compared

against. The metric itself has no argument in which a time could appear, so a temporal discrepancy is scored as displacement. Neither side of the comparison is a state at an instant. A predictor computes $\hat{s}$ from a window of sensor data. A label is produced by an exposure, an annotator or tracker working on the exposed image, and a geometric transfer into another sensor’s frame.

The predictor side of that mismatch is fixed by a released configuration and is measurable from released artifacts. The event branch of the detector examined here [23] consumes a window of 50 ms divided into ten bins, and the preprocessing script of its own repository builds the window boundaries by counting backwards from label timestamps, so each window ends at the label it is scored against (Sec. 3.2). Under uniform weighting the evidence in such a window is centered 25 ms before the instant the label asserts. Whether the trained network weights its bins uniformly is a separate question, and occluding each bin in turn on real validation samples answers it: the influence-weighted centroid is -23.81 ms (Sec. 3.3). A network whose newest window sits there and whose output is scored at the label instant extrapolates over that interval.

The training objective already places the output at the label instant, and the measurement of Sec. 3.4 is consistent with it, at -2.40 ± 7.18 ms, which is a failure to reject and not a demonstration that they coincide. The objective does not fix the interval that measurement leaves open. That window’s centroid and the measured output time are 26.2 ms apart, so the output is an extrapolation over that interval. The velocity that measurement regresses on is finite differenced from the same labels the residual is measured against, and a forward difference shares their noise, returning an offset from a detector that has none. The estimator is therefore calibrated on synthetic tracks before it is used, and carries a placebo term on which no lag can load.

The second measurement concerns the same class of undeclared timing property on the label side. DSEC publishes each frame’s own exposure interval, and across the eighteen training sequences whose exposure files resolve, the width varies by a factor of 127 against a frame period fixed at 50 ms, with 32.5 % of frames pinned at an auto-exposure ceiling of $14996 \mu\text{s}$ in six sequences (Fig. 1). Those six carry the harmonic series of a 100.00 Hz mains flicker, with nothing at the odd multiples of 50 Hz, and between 95.4 % and 100.0 % of labeled boxes carry more

power at the fundamental than at an off-frequency arm on the same events (Sec. 4). A $10\,000\,\mu\text{s}$ period inside a $14\,996\,\mu\text{s}$ exposure displaces event times with no motion involved, and Sec. 4.4 reports what that leaves of a per-object statistic.

The single timestamp the metric is indexed by does not specify the temporal support of either side of the comparison. Three quantities are collapsed into it, and this work separates and measures each on released artifacts: the evidence support of a released recurrent event detector, whose newest window is centered 23.81 ms before the label instant and carries 22.7 % of its influence, the rest reaching past the second measured (Secs. 3.2–3.3); the output time of the same checkpoint, identified by a vector regression with a placebo term and a sign-flip control on the velocity estimator, which places it at the label instant and 26.2 ms from that window (Secs. 3.4–3.5); and the exposure-conditioned support of DSEC’s labels, where the harmonic series of a 100.00 Hz mains flicker in six ceiling-exposure sequences bounds what a per-object temporal statistic measured inside one published exposure window can be attributed to (Sec. 4).

2 Related work

Recovering a time offset from image-plane displacement. Qin and Shen [5] calibrate a clock offset online by modeling a feature observation as $z(t_d) = [u, v]^T + t_d V$, with V differenced from consecutive observations. That is the estimator of Sec. 3.4, published in visual-inertial odometry, and the same work identifies the offset by making one scalar reconcile two channels. Temporal calibration removes the offset between two *sensor* clocks by pooling over features, so the spread of per-feature estimates is its error bar. Here the offset lies between a predictor’s clock and a *benchmark’s* label clock. Streaming perception [6] folds wall-clock compute latency into average precision and re-anchors any detector from box coordinates alone; later work adapts to the measured runtime delay [35], a continuous-stream framework scores the latest available output against high-rate ground truth and reports reordered rankings [29], and temporal tolerance has been argued for time-series event detection [36]. Each of those latencies is the time from an event to an output and falls as the device gets faster. Sensor latency, the illumination-dependent delay between a change and the timestamp the sensor assigns it, has also been measured and corrected [28], at microseconds, and nighttime event imaging reports illumination-dependent temporal trailing which it corrects by learning a per-event timestamp calibration [24]. Those corrections target the sensor; here the released exposure metadata and event stream bound what a per-object temporal statistic computed inside one pub-

lished exposure can be attributed to. Inside the citation graph of LET-3D-AP [1], a latency-aware average precision scores 3D detection at the instant a prediction becomes available on a named device, on nuPlan [37], which is again compute latency. The interval measured in Sec. 3.5 is fixed by a data representation and does not. The event-RGB frequency mismatch is named in [38], which scores with unmodified average precision.

Decomposing localization error along a privileged axis.

LET-3D-AP [1] decomposes camera-only 3D localization error along the line of sight and defines an average precision tolerant of the longitudinal part, and reports a camera-to-label synchronization gap as a per-dataset interval that depends on horizontal object position. Sec. 3.4 keeps the idea of a privileged axis but replaces the sensor’s line of sight with the object’s own velocity vector, and estimates a time rather than defining a tolerant score. TIDE [7, 8] and DETAD [9] bucket detection error atemporally, so a temporal offset falls inside their localization bucket, HOTA [10] takes the label clock as exact, and the response of average precision to isotropic box perturbation [11] supplies the isotropic null the supplement uses.

Per-object time in labels. Waymo’s `camera_synced_box` solves t_{capture} per object and stores the resulting box shift rather than the time [2, 3]. Nuñez et al. [4] measure within-frame per-object annotation offsets on Argoverse 2 and TruckScenes and define a dispersion of the *spatial* error. Each computes a per-object time in order to remove it. The event-vision label formats carry the field: the `Prophesee dtype` used by Gen1, the 1 Mpx dataset and DSEC-Det has a per-box τ . What no release fills is that field’s per-object degree of freedom. Across the 60 released DSEC-Det sequences, 390 118 boxes take 70 379 distinct values of τ , 5.54 boxes per value, and those values are the 50 ms frame clock.

Event benchmarks and their exposures. Gen1 [16] labels are drawn on gray-level images assembled from asynchronous per-pixel measurements. On the 1 Mpx dataset [17] the event count inside a labeled box at its own timestamp is often too small to support detection [18]. DSEC-Det [13] labels by tracking [14] on rectified frames and warps into the event view, and Ev-3DOD [19] interpolates lower-rate annotations into high-rate 3D labels. That a frame’s exposure is long relative to event timing has been stated. REFID [20] argues that accumulating events from a single frame timestamp loses information, deblurring work names auto-exposure as why the width is unknown [21], and RENet [22] uses a DSEC exposure interval as its event window while describing that exposure as short. None of these reads the released per-frame widths or measures

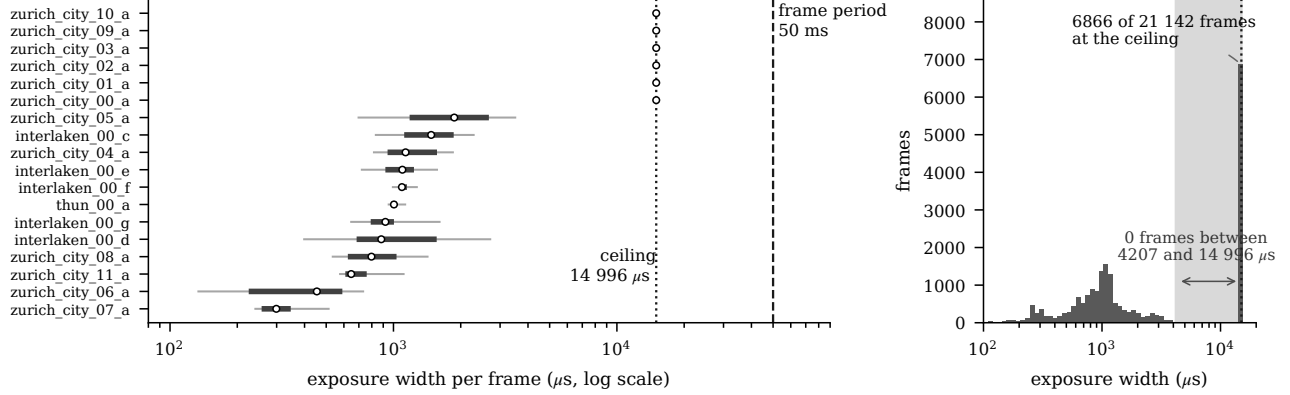


Figure 1: Published per-frame exposure width across the eighteen DSEC training sequences whose exposure files resolve. Left: per sequence, the median with the interquartile range and the 5–95 % band, on a log axis, against the fixed 50 ms frame period. Right: the pooled distribution over 21 142 frames. 32.5 % of frames sit at the auto-exposure ceiling of 14 996 μ s, and zero frames fall between the largest non-ceiling width of 4207 μ s and that ceiling, so the width is a two-level factor on this data rather than a swept one. The six sequences that reach the ceiling are the ones measured in Sec. 4.

what is inside one. State space backbones [25] removed by architecture the dependence of an event model’s score on inference-time window duration, and adaptive spatial-temporal windows [27] retrain a detector under each of nine partitioning strategies and report the score of each. Both treat the window as a design variable to choose. Here it is a released fact to locate. That the released window is causal is already in print: LEOD [26] states that event detectors take only events triggered before the query time, and builds a time-flip augmentation on the consequence that forward inference favors receding objects and backward inference approaching ones. Where inside that window the evidence sits is not measured there. Protocol work supplies the standards these measurements are held to [31–34]: a repaired protocol can reverse a field’s conclusion, a reported score is an estimate with a variance, and a label audit is judged by whether what it finds moves benchmark results.

3 The interval between evidence and label

3.1 Definition

Let a predictor P emit a state at query time t with normalized temporal support kernel $k_P(u)$, the share of the emitted state’s information coming from $t + u$, of centroid c_P . A benchmark scoring at a nominal instant treats that centroid as zero. Of the benchmarks and released detectors whose documentation we read, Gen1, 1 Mpx, DSEC-Det, DSEC-3DOD, DSEC-Flow, MVSEC, the RVT release and the SSM-ViT release [25], none states or checks it.

The *effective timestamp* τ_P is not c_P . For a predictor linear in position measurements at bin centers c_k with

weights a_k summing to one, the conditional expectation of the output under constant velocity is the true position at $\tau_P = \sum_k a_k c_k$. Non-negative weights give $\tau_P = c_P$, while a predictor that uses older evidence to compensate its own lag has negative weights and moves τ_P toward zero without moving c_P . Sections 3.2 and 3.3 measure the newest window’s contribution to c_P , Sec. 3.4 measures τ_P , and Sec. 3.5 reports the interval between them.

3.2 The window and the instant it is indexed by

Two released files fix the input side. The dataset configuration `config/dataset/gen1.yaml` of the detector’s own repository [23] names the representation `stacked_histogram.dt=50_nbins=10`, a window of 50 ms divided into ten bins of 5 ms. Line 405 of `scripts/genx/preprocess_dataset.py` constructs the array `ev_repr_timestamps_us_end` by counting backwards from a label timestamp in steps of the window duration, and extends it between consecutive label timestamps. Each representation is therefore indexed by the end of its window, and those ends are placed on label times. A window attached to a label at time t holds the events of $[t - 50 \text{ ms}, t]$, whose uniform weighting centroid falls 25 ms before the instant t at which the label asserts a state. Both statements are read from the released source rather than from a description of it, and the 25 ms figure is a reference point rather than a property of the trained network.

3.3 The measured influence profile

τ_P is defined on influence weights, which are measurable for a released checkpoint. We take 480 samples of the

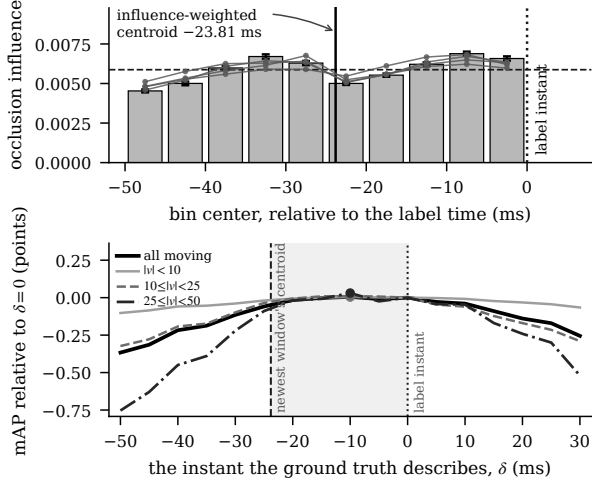


Figure 2: (a) Occlusion influence of each of the ten bins of the released `rvt-t` input window, over 480 real Gen1 validation samples with the recurrent state warmed eight steps, with the standard error of each mean. The dashed line is the mean over bins, the dotted line the label instant, and the solid line the influence-weighted centroid; grey lines are the 4 other checkpoints, each rescaled to its own mean. (b) Average precision against the instant δ the ground truth is displaced to describe, by object speed in px/s, each curve relative to its own value at the label instant, with its maximum marked. The shaded band is the interval between the newest window’s influence centroid and the label instant. The curves deepen with speed as the overlap arithmetic requires, and all of them peak between the two lines.

released Gen1 validation split with the released `rvt-t` checkpoint, warm the recurrent state for eight steps, and occlude each bin in turn, reading a bin’s influence as the relative L_2 change its occlusion produces in the emitted detection tensor. The recurrent state given to the occluded pass is the state the unoccluded one received, and the state it returns is discarded, so the profile is a property of the current window and not of a perturbed history.

The profile varies across bins and is balanced between halves (Fig. 2a): the coefficient of variation across the ten bins is 0.131, the heaviest bin carries $1.52\times$ the lightest, and the five older bins hold 48.6 % of the influence against 51.4 % for the five newer. The influence-weighted centroid is therefore -23.81 ms against the uniform-weight value of -25.00 ms, a difference of $+1.19$ ms or 23.8 % of one bin width. For this configuration the centroid of the input is the uniform-weight centroid of the window to within a fraction of one bin, and the configuration fact of Sec. 3.2 is also a measured property of the trained network. This is the current window’s contribution to k_P , taken with the recurrent state held at the value entering the step, and not k_P itself. Zeroing a bin also pushes the input off the data manifold, so the fill is varied and the instrument replaced. Setting a bin to its dataset mean gives -26.35 ms and taking it from an unrelated sample -25.01 ms; a gradient sensitivity, which occludes nothing at all, gives -22.35 ms. The

four centroids span 4.01 ms against a bootstrap standard error of 0.054 ms on the zero-fill centroid, so the spread is the choice of instrument. Occlusion still measures a sensitivity, which equals the linear weight a_k only for a predictor linear in its bins.

The rest of the kernel. The detector is recurrent, so k_P does not end at the window. Restoring the state saved before the step at $t - 50L$, occluding that whole window and letting the state propagate to t measures each past window’s influence over 576 samples. The newest carries 22.7 % of the total over a 1000 ms horizon, half is reached only by -175 ms, and zeroing the state changes the output by 0.0867 of its norm against 0.0231 for removing the newest window entirely. The tail does not decay to zero within that second, so the influence-weighted centroid follows the horizon instead of converging: -161 ms over 500 ms of history and -299 ms over 1000 ms. Filling a past window from an unrelated sample rather than with zeros deepens the plateau, the decay across the second falling from 10.3 to 2.2. This predictor’s support therefore has no centroid the measured horizon identifies.

Across the released checkpoints and a second architecture. RVT releases three Gen1 checkpoints differing only in embedding width, attention head dimension and FPN depth. SSM-ViT [25] releases two more in which each stage’s ConvLSTM is replaced by an S5 state space layer, at 9.68 and 18.19 M parameters against RVT’s 9.87 and 18.54; it distributes RVT’s own preprocessed Gen1 and inherits its window construction. On the same sequences, and driving each the way its own release drives it, all 5 checkpoints put the newest window’s centroid between -23.76 and -24.72 ms, none further than 1.24 ms, or 25 % of one bin, from the uniform-weight value. The state the SSM release carries between evaluation chunks reaches none of its outputs, which bounds that model’s support at the 21-window chunk its evaluation steps through; the supplement measures it. The support behind that window is not fixed in the same way either: over 576 samples of the three RVT capacities, the newest window’s share of a 1000 ms horizon falls from 22.7 % to 19.0 % and the support centroid moves from -299 to -339 ms.

3.4 The output time

The influence profile is a statement about the network’s input. Its output can carry a different time, since a network that weights its window evenly can still emit a state describing a later instant. That quantity is measured from the same checkpoint’s own detections.

A detector reporting the state an object occupied τ earlier displaces its prediction by $-\tau v^*$, where v^* is the label

velocity, so $\tau = -\tau_P$ on the axis of Sec. 3. That is a statement about a vector, and it is estimated as one. With $\hat{c}$ the predicted center, c^* the label center, $\hat{u} = v^*/\|v^*\|$ and R a quarter turn,

$$\hat{c} - c^* = b - \tau v^* + \rho Rv^* + \gamma\sqrt{A}\hat{u} + \varepsilon \quad (1)$$

is fitted by stacking both components of every match into one least squares problem, with standard errors clustered by sequence. It is the temporal calibration of [5] carried to a box center, taken as a slope rather than as the per-object ratio $-\langle \hat{c} - c^*, \hat{u} \rangle / \|v^*\|$, whose variance diverges as its denominator approaches zero. Fitted on these rows its mean is $+24.22 \pm 27.73$ ms, on the very hypothesis Sec. 3.5 excludes, with an interquartile range of 430 ms. Written this way τ is identified by construction under the model: no magnitude of the residual is taken, and τ does not load on a direction estimated from the same noisy velocity.

The remaining two terms play different roles. Rv^* is the label velocity rotated a quarter turn. It has the same magnitude and the same measurement noise as v^* , and no temporal offset can load on it, so ρ is a placebo whose null distribution the estimator supplies for itself. $\sqrt{A}\hat{u}$ absorbs a displacement along travel proportional to box side, which is needed because that residual grows with box size and box size correlates with label speed at $+0.139$ on this split.

The finite difference used for the velocity. v^* is finite differenced from labels, and the choice of difference changes the answer. A forward difference $(c_{k+1}^* - c_k^*)/\Delta$ carries the label noise δ_k of the current center with weight $-1/\Delta$, and the residual $\hat{c} - c_k^*$ carries the same δ_k with weight -1 ; their covariance is $\text{Var}(\delta)/\Delta$ when τ is exactly zero. A backward difference carries δ_k with the opposite sign and produces the opposite artifact. The centered difference $(c_{k+1}^* - c_{k-1}^*)/(\Delta_f + \Delta_b)$ does not use c_k^* and is free of it. Fitting all three is therefore a control on the estimator itself, and it is calibrated before use. On 2000 synthetic tracks whose detector has no lag whatever, with label centers corrupted at the dispersion the supplement’s difference ladder measures on Gen1 (0.6515 px), the forward estimator returns -5.64 ms and the backward one $+5.84$ ms against a closed form of -5.63 ms, they sum to $+0.20$ ms, and removing the label noise returns all three to $+0.02$ ms. The same simulation recovers an injected lag at 0.92 of its size, the attenuation an errors-in-variables treatment predicts at 0.933 for this noise.

The measurement. The released checkpoint is run over the Gen1 validation split with its recurrent state carried across steps, and its detections at each labeled frame are matched to the ground truth at an overlap of 0.3 or above. Requiring both label neighbors, so that a centered velocity exists, and discarding the matches whose two neigh-

Specification	τ (ms)	placebo ρ (ms)
<i>velocity estimator, no geometric controls</i>		
forward difference	-14.68 ± 4.99	-6.53 ± 1.68
backward difference	$+45.65 \pm 5.59$	-6.15 ± 1.71
centered difference	$+17.51 \pm 5.06$	-7.76 ± 1.92
<i>controls added, centered difference throughout</i>		
+ box side along travel	$+14.62 \pm 5.47$	-7.75 ± 1.91
+ four geometric terms	-7.62 ± 7.01	-4.12 ± 1.90
+ per-sequence bias	-2.40 ± 7.18	-2.66 ± 1.71

Table 1: The output-time offset τ of the released `rvt-t` checkpoint over 22 534 matched boxes of 376 Gen1 validation sequences, and the placebo ρ beside it. ρ is the coefficient on the label velocity rotated a quarter turn, which no temporal offset can reach, so it is a null the same data supplies. Standard errors are clustered by sequence. Under the full control set the forward and backward estimators are still 35.0 ms apart and straddle the centered one, as the shared label-noise term predicts.

bors coincide on the integer label lattice, leaves 22 534 of 35 523 over 376 sequences at a median overlap of 0.873. Gen1’s `track_id` is a per-frame index rather than a persistent identifier, so association is rebuilt by same-class overlap between adjacent label times. No speed floor is imposed.

Table 1 is a specification ladder, and the specification was chosen by the placebo rather than by τ . Under Eq. 1 with only the box-side term, $\tau = +14.62 \pm 5.47$ ms, but ρ is -7.75 ms at 4.0 standard errors, which indicates a term omitted from Eq. 1 that Rv^* is a proxy for. A split by the sign of the horizontal velocity locates it: ρ changes sign across that split, at -27.99 ms and $+21.84$ ms, which a rotational term would not do. Most label motion here is horizontal, so Rv^* maps a horizontal speed onto a vertical residual, where a bias differing between oncoming and same-direction traffic lands with nothing rotational present. Entering box height and image row along $\hat{y}$ and box width and image column along $\hat{x}$, and then a constant vector per sequence, returns the placebo to -2.66 ms at 1.6 standard errors. That is the specification reported, and the criterion that selected it is a coefficient on a regressor orthogonal to v^* .

Under it, $\tau = -2.40 \pm 7.18$ ms, which is 0.33 standard errors from the label instant and 3.65 standard errors from the influence centroid of Sec. 3.3; a cluster bootstrap over sequences puts the 95 % interval at $[-17.0, +11.5]$ ms. The controls do not absorb a lag and the axis pairing does not make the answer: an injected offset is recovered at 1.000 of its size under the full control set, entering all four geometric terms on both axes gives -3.62 ± 7.41 ms, and across every specification carrying geometric controls τ spans -20.2 to $+11.1$ ms.

3.5 The interval between evidence and output

The two measurements are of one architecture at one configuration. Its newest input window is centered 23.81 ms before the label time (Sec. 3.3) and its measured output time is consistent with that label time (Sec. 3.4). The measurement does establish the separation: the output is incompatible with that window’s centroid at 3.65 standard errors, and 26.2 ± 7.2 ms is the point estimate of their distance under the reported specification rather than a latency of this architecture. On the convention of Eq. 1 the centroid hypothesis is $\tau = +23.81$ ms, and no specification carrying geometric controls reaches it: τ spans -20.2 to $+11.1$ ms across them. The rest of the support lies further back, so 23.81 ms is the smallest interval consistent with the measurement rather than its value. The standard error quoted is the output-time term’s; the influence centroid contributes 0.054 ms of sampling error and 4.01 ms of dependence on the instrument (Sec. 3.3), and the specification spread above is wider than either.

That outcome is the one the training objective enforces: the detector is trained against the ground truth at the label time on a window ending there, so a measured offset near zero reports the objective. The objective does not enforce the interval, and the instant the output describes and the centroid of the evidence used to produce it are different quantities that one benchmark timestamp stands for. At the ninetieth percentile of label speed on this split, 25.7 px/s, the 23.81 ms is 0.61 px of displacement, comparable to the annotation dispersion of the labels themselves (0.6353 px), and at the ninety-ninth it is 1.29 px. That window’s centroid and the output time are both measurable from released artifacts alone.

The interval on the benchmark’s own metric. The benchmark can be asked in its own units: displacing the ground truth of every labeled frame by δv^* scores the released predictions against the state each object occupied δ later and leaves the detector untouched; a detection carries no velocity of its own. Over 20 296 frames and 40 698 boxes, 68.7 % of which carry a centered velocity and the rest are ignored, mAP against δ is smooth with its maximum at -10 ms (Fig. 2b). On the axis of Eq. 1 that is $\tau = +10$ ms, on a top flat to 0.04 points across the interval. A bootstrap over frames puts that argmax between -20 and 0 ms in 95 % of 1000 resamples, and in none of them does it reach that window’s centroid, so it identifies no value.

The flatness of that curve is the second result. The whole sweep moves mAP by 0.9 points and displacing the ground truth to that window’s centroid costs 0.04 points, 0.12 % of 0.3346: the separation measured above is real and this metric does not register it. Nor does it on any of the 5 re-

leased checkpoints: displacing each to the instant the metric prefers for it leaves their ordering unchanged in four speed strata, and cannot change it, since within a stratum the largest gain from alignment is 52 % of the smallest gap between adjacent models (supplement). A displacement costs overlap in proportion to $\tau \|v^*\|$ against the box side, and the median label speed here is 4.0 px/s on a 41.5 px box, so 23.81 ms is 0.10 px. Sorted by speed the sweep deepens as that requires, to 0.78 points between 25 and 50 px/s, then shallows again above 50, where mAP is 0.139 against 0.286 below it and objects are lost to detection rather than to localization.

4 A mains signature in DSEC’s ceiling-exposure sequences

4.1 Exposure and labels

The predictor side of the mismatch is fixed by a window; the label side is fixed by an exposure. DSEC publishes per-frame exposure metadata, and the files resolve for eighteen train sequences. The width varies by a factor of 127 against a frame period fixed at 50 ms, and within one recording by up to a factor of 12.5. The auto-exposure ceiling of 14 996 μ s holds 32.5 % of frames, in six of the eighteen sequences, and the distribution is not continuous over that range: of the 21 142 frames, zero fall between 4207 and 14 996 μ s (Fig. 1).

The released DSEC-Det labels comprise 390 118 boxes on 6693 tracks across 60 sequences, and every label sits on the frame clock. The median spacing between consecutive label times is 50 001 μ s and the fraction shorter than four fifths of a frame period is 0.00 %. Over 361 628 triples of consecutive labels on a track the median distance from the middle box center to the linear interpolant of its neighbors is 0.707 px and 7.1 % fall below a hundredth of a pixel, where interpolated labels would place that fraction near unity, so no construction term from interpolation arises within this archive. DSEC-Det ships more than one annotation version [15]; the supplement names the one measured here.

The label center carries noise, and that noise is measurable on the labels themselves. On a uniform grid the variance of the k -th difference of a track’s box center is $\binom{2k}{k} \sigma_c^2$ plus a motion term whose order rises with k , so $\text{Var}(D^k x) / \binom{2k}{k}$ approaches σ_c^2 from above, and halving the sampling rate multiplies the motion term by 4^k and leaves σ_c untouched. Over 5250 DSEC-Det tracks the two routes agree from $k = 4$ upward and the ladder extrapolates to $\sigma_c = 0.4974$ px, of which the 0.5 px integer lattice the release stores box centers on contributes 0.1443 px. Median box-center image speed is 40 px/s, with 360 px/s at the ninety-ninth percentile.

4.2 The line, and the per-pixel test

Swiss mains is 50 Hz, so lamp intensity flickers at 100.00 Hz. An event camera responds to log-intensity change rather than to intensity, so both edges of that oscillation produce events and the arrival rate carries the harmonic series. Sweeping 20–450 Hz on each whole recording, with no band fixed in advance and each frequency divided by the median of its own neighbourhood, returns that series in every ceiling sequence: the ceiling median is 30 at 100.00 Hz, 125 at 200 Hz and 15 at 300 Hz, and the second harmonic is the strongest line in four of the six, reaching 2518 on `zurich_city_09_a` and never falling below 14.7. At the odd multiples of 50 Hz there is nothing: 1.8, 4.5 and 1.6 at 50, 150 and 250 Hz. That absence is what a rectified supply predicts and what a periodic source of any other origin need not obey. The daytime sequence carries the fundamental at 0.5 and the second harmonic at 6.2, against 14.7 for the weakest ceiling sequence.

Per pixel, the Rayleigh statistic $Z = R^2/N$, for R the resultant length of the event phases at the test frequency, is $\text{Exp}(1)$ under independent uniform phases, so its null has median 0.693, ninety-fifth percentile 3.00 and a rate of 0.1 % at $p < 10^{-3}$. Independence is a claim about the data and it is checked: over 35 990 pixels of the ceiling sequence with at least 100 events, the same estimator on the same events at 137 Hz returns 0.581 with a $p < 10^{-3}$ rate of 0.04 %, and on 17 800 daytime pixels at 100.00 Hz it returns 0.719 at 0.08 %, so per pixel the reference distribution is not exceeded off the line or away from the lamps. At 100.00 Hz on the ceiling sequence the median Z is 2.935 and 24.07 % of those live pixels reach $p < 10^{-3}$. DSEC’s own description attributes high night event rates near street lamps to flashing lights [12], and event noise is known to rise in dim light [30]. Neither is quantified per pixel in the release.

4.3 All six ceiling sequences, per labeled box

The per-pixel test above is one sequence against one daytime control. The same statistic taken per labeled box, over the events inside that box during that frame’s own exposure window, extends to every sequence that reaches the ceiling. Table 2 reports all six of them over 16 516 boxes. The median per-box Rayleigh statistic at 100.00 Hz runs from 68.4 to 310.9 with a median over sequences of 114.58, against 4.01 for the same estimator at 137 Hz on the same events.

The analytic null does not transfer from a pixel to a box, since 4.01 is not 0.693, and over eight off frequencies the pooled median is 38.4, 55 times the analytic one, since events inside one box during a 14 996 μs exposure arrive in bursts. No per-box p -value is quoted, and 137 Hz is the most favorable of them, the line standing at 4.0 times their pooled median. A ratio needs no null: dividing Z at the

Sequence	boxes	m	$Z_{100.00}$	Z_{137}	locked
<code>zurich_city_00_a</code>	2914	0.428	117.6	3.15	100.00 %
<code>zurich_city_01_a</code>	2134	0.440	99.1	4.69	99.63 %
<code>zurich_city_02_a</code>	619	0.445	68.4	2.75	99.68 %
<code>zurich_city_03_a</code>	241	0.489	310.9	25.85	95.44 %
<code>zurich_city_09_a</code>	6997	0.466	251.5	7.82	99.23 %
<code>zurich_city_10_a</code>	3611	0.438	111.6	3.33	100.00 %

Table 2: Every DSEC training sequence whose exposure is pinned at 14 996 μs , measured per labeled box over the events inside that box during that frame’s own published exposure window, with at least 400 events per box. m is the median over boxes of the modulation depth $m = 2|C|$ at 100.00 Hz, against a floor of 0.089 for 400 events drawn at random phase. $Z_{100.00}$ and Z_{137} are the median over boxes of the Rayleigh statistic at 100.00 Hz and at the off-frequency control on the same events. The analytic $\text{Exp}(1)$ median of 0.693 does not describe either column, which is why the two are compared with each other; *locked* is the fraction of boxes whose local excess at 100.00 Hz exceeds the ninety-ninth percentile of the same quantity at 137 Hz.

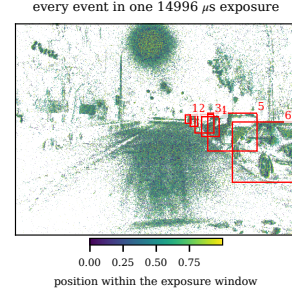


Figure 3: Every event inside one published 14 996 μs exposure window of `zurich_city_09_a`, colored by its position within that window, with that frame’s released boxes numbered left to right. The frame is the median-dispersion one of 602 candidates, not the largest.

line by the median of $Z(f)$ over each box’s own neighbouring frequencies gives 1.87 over 16 516 boxes against 0.11 for the same construction at 137 Hz, and 99.63 % of boxes exceed that arm’s ninety-ninth percentile. One window is transform-limited to 67 Hz, so a box establishes that the modulation reaches it and not which harmonic carries it; the identification is the sweep of Sec. 4.2.

4.4 A per-object temporal statistic under that signature

That signature bounds any temporal statistic defined per object inside one published exposure window, and the bound is measurable. Let a labeled frame have published exposure $[a, b]$ of width $w = b - a$ and mid-exposure $m = (a + b)/2$, and let box i contain the events $\{(x, y, t)\}$ falling inside it with $t \in [a, b]$. The *event centroid* of the box is

$$\bar{t}_i = \frac{1}{N_i} \sum_{k \in i} t_k - m, \quad (2)$$

the mean event time referred to the instant the dataset indexes the frame with. If the evidence inside each box were uniform over $[a, b]$ then $\bar{t}_i$ would have variance $w^2/12N_i$, so the excess over that null is

$$\sigma_{\text{evt,excess}}^2 = \text{Var}_i(\bar{t}_i) - \frac{1}{n} \sum_i \frac{w^2}{12N_i}. \quad (3)$$

The subtrahend holds for arrivals independent and uniform on $[a, b]$, which these are not. Reassigning a frame’s box events among its boxes at random, which leaves every structure common to the frame intact and destroys only box identity, gives a null of 78.6 μs , 0.85 times the analytic one rather than above it, and an excess of 191.1 μs ; the supplement gives that construction. A box enters with at least 200 events and a frame with at least 3 qualifying objects. On `zurich_city_09-a`, 1134 of 1557 labeled frames qualify, with a median of 6 objects each. The within-frame standard deviation of $\bar{t}_i$ is 206.6 μs against an analytic null of 93.0 μs , the excess is 183.6 μs , and the standard deviation exceeds that frame’s own null on 90.1 % of frames (Fig. 3). On `interlaken_00-c`, where the exposure is 1556 μs and the 200-event floor leaves 20 usable frames, the standard deviation is 18.0 μs against a null of 21.6 μs and the excess is zero.

The supplement tabulates what that dispersion can be attributed to. Boxes of the same shapes at random positions give 306.3 μs , above the annotated boxes’, so it is a property of where evidence falls during the exposure and not of objecthood. Removing the 14 334 pixels that reach $p < 10^{-3}$ per pixel changes it to 180.6 μs and a matched random exclusion to 182.9 μs , so neither exclusion moves it, and the median Z over all live pixels, 2.935 against 0.693, says why: a weakly modulated population remains in the clean arm.

That population is reached without a mask. Recomputing the dispersion inside quartiles of each box’s own modulation depth $m = 2|C|$ at 100.00 Hz, over 6841 boxes of 1134 frames, the excess rises monotonically with m from 93.1 to 208.0 μs . The same procedure at 137 Hz, where the line is absent, also rises from 150.3 to 187.6 μs , so part of the trend belongs to stratifying on any amplitude statistic; the 100.00 Hz span is the wider and its lowest stratum lies below the control’s lowest. An unmodulated stratum would separate the two and none exists: m has a tenth percentile of 0.355 against a floor of 0.125. The dispersion is therefore a property of the event stream inside the published window that this data does not resolve into a scene component and an illumination component. That is an identifiability bound and not a timing error: a per-object temporal statistic measured inside one of these exposures cannot be attributed to object timing until the illumination-driven structure is separated from it.

That dispersion bounds the precision to which one scalar timestamp denotes an instant, and it converts to the scale

a benchmark scores on through the object speeds above. A dispersion of 183.6 μs , whatever its source, displaces a box center by 0.0073 px at the median object and 0.0661 px at the ninety-ninth percentile, 1.5–13.3 % of σ_c ; reaching σ_c would require 2709 px/s, which DSEC-Det does not contain. The frame-common offsets are smaller still, -16.8 and -102.3 μs from the mid-exposure instant over all events and over the labeled subset. Against the dispersion, the largest of the three label-side terms, the newest window’s influence centroid of Sec. 3.3, measured on a different dataset, is larger by a factor of 130, and of the three it is the one a declared timestamp would remove.

5 Limitations

A per-object temporal claim on these sequences is bounded rather than identified. The dispersion rises with each box’s own modulation depth, no unmodulated stratum exists in any of the six, and the one spatial statistic chosen for its immunity to a spatially uniform source does not survive its own background control: an equal-area annulus of each box’s surroundings aligns at least as well as the object in four of them. That alignment is ego-motion, which this data does not separate from a per-object event centroid. The narrow-exposure contrast is one sequence, so exposure width and illumination move together in it, and a denoiser ablation against the noise of [30] is not run.

On the predictor side, everything is one dataset and 5 released checkpoints of two architectures that share a repository, and occlusion reports a sensitivity that equals a linear weight only for a predictor linear in its bins. The support measurement weights windows by the norm of a difference and neither fill is the training distribution, so its shares are a ranking and its centroid is horizon-dependent. The output-time measurement is a null bounded by its precision: it excludes an offset as large as the newest window’s centroid at 3.65 standard errors, says nothing about one of a few milliseconds, and assumes constant velocity across the label neighbourhood its difference is taken over. Its specification was chosen by a coefficient orthogonal to v^* , though the step that returned it to null also moved τ .

6 Conclusion

A detector’s newest input window is centered 23.81 ms before the instant it is scored at, within 1.24 ms across 5 released checkpoints of two architectures; most of its influence lies further back; and its output time is bounded away from that window at 3.65 standard errors under every geometrically controlled specification, a separation average precision registers as 0.04 points and no reordering; and DSEC’s six ceiling exposures bound what a per-object statistic inside one can be attributed to.

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